Text-to-Video Diffusion with ZeroScope v2: Pipeline Characterization and Weakness Analysis

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***Abstract*—We characterize an open-source latent video diffusion baseline, ZeroScope v2 (576w) [2], and quantify three weaknesses that limit short-clip generation quality on a single consumer-grade T4 GPU. The pipeline pairs a 3D U-Net with the DPM-Solver multistep scheduler [7] and is driven by CLIP [6] text embeddings under classifier-free guidance [5]. On a single complex urban scene generated at 576×320 with 24 frames, 40 denoising steps, and guidance scale w=7.5, the baseline reaches CLIP-SIM = 0.3241, LPIPS-Temp = 0.0687, SSIM = 0.6745 and PSNR = 19.48 dB between consecutive frames. We frame these numbers around three diagnoses: (W1) inter-frame perceptual drift consistent with the limited effective range of temporal self-attention; (W2) semantic variance across the clip that we attribute to the 77-token CLIP context bound; (W3) high inter-frame variance under a 40-step DPM-Solver budget. The diagnoses motivate the targeted interventions explored in our companion Phase 2 work [14].**

***Index Terms*—Latent video diffusion, ZeroScope, DDPM, 3D U-Net, CLIP, classifier-free guidance, temporal coherence.**

I. Introduction

Text-conditioned video generation has progressed from frame-by-frame image synthesis to genuinely spatiotemporal models, but the short-clip regime on commodity hardware remains brittle. The visible failure modes are familiar to anyone who has run a diffusion sampler: the model agrees on *what* a scene is, then disagrees with itself about how it should evolve. This paper does not propose a new architecture. Its purpose is to fix the baseline well enough that the follow-on enhancements [14] have a meaningful comparison point.

We use ZeroScope v2 (576w) [2], a publicly-available latent video diffusion model built on the standard LDM recipe [1] and shipped with a 3D U-Net. It is small enough to run end-to-end on a single NVIDIA T4 in Google Colab, which constrains every design choice we make: 24 frames, 576×320 pixel output, half-precision weights, and chunked VAE decoding.

Contributions. (1) A worked mathematical account of the diffusion objective used by ZeroScope, with the exact prediction target and sampler that the pipeline implements; (2) an architectural description of the 3D U-Net and its three attention modes, traced through to the executed code; (3) a quantitative baseline of the pipeline on four metrics, against which the weaknesses are concrete rather than rhetorical; and (4) three weakness diagnoses (temporal drift, semantic variance, sampling-induced inter-frame variance) with the metric trace that backs each one. Section II fixes notation; Sec. III walks through the architecture; Sec. IV states the experimental setup; Sec. V presents the measurements and root-cause hypotheses; Sec. VI discusses the limits of the analysis itself.

II. Background and Mathematical Foundation

A. Diffusion process (DDPM)

DDPM [3] defines a forward Markov chain that gradually corrupts data $\mathbf{x}_0$ into Gaussian noise:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\big(\mathbf{x}_t;\ \sqrt{1-\beta_t}\,\mathbf{x}_{t-1},\ \beta_t\mathbf{I}\big). \tag{1}$$

A standard reparameterization permits sampling $\mathbf{x}_t$ in closed form from $\mathbf{x}_0$,

$$\mathbf{x}_t = \sqrt{\bar\alpha_t}\,\mathbf{x}_0 + \sqrt{1-\bar\alpha_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}), \tag{2}$$

with $\bar\alpha_t = \prod_{i=1}^{t}(1-\beta_i)$. The reverse chain $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is parameterized through a noise predictor $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$; for video, the denoising step is applied to the full latent volume simultaneously, not frame by frame.

B. Score interpretation

Up to a known scaling, the noise predictor coincides with the score of the perturbed distribution [4]:

$$\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \ \approx\ -\sigma_t\,\nabla_{\mathbf{x}} \log p_{\sigma_t}(\mathbf{x}_t). \tag{3}$$

Learning to predict noise is therefore learning a tractable score estimator, which is what makes the sampler at inference time a deterministic ODE solve.

C. Training objective

Maximising the ELBO of (1)-(2) reduces to the simple regression target [3]:

$$\mathcal{L} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\big\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\big\|^2\right]. \tag{4}$$

This is the ϵ-prediction objective, which ZeroScope inherits from the underlying LDM [1]. Two alternatives in common use, $\mathbf{x}_0$-prediction and v-prediction with $\mathbf{v} = \sqrt{\bar\alpha_t}\,\boldsymbol{\epsilon} - \sqrt{1-\bar\alpha_t}\,\mathbf{x}_0$ [10], trade sharpness for stability differently; for compatibility with the DPM-Solver sampler we use, ϵ-prediction is the right choice [7].

D. Noise schedule

The schedule $\{\beta_t\}$ governs how aggressively noise is injected. A linear schedule front-loads noise and is known to destabilize high-resolution training [9]; a cosine-like schedule keeps $\bar\alpha_t$ smooth near the endpoints. Our DPM-Solver multistep scheduler uses a cosine-style profile inherited from the pre-trained model.

E. Classifier-free guidance

At sampling time, the conditional and unconditional predictions are blended [5]:

$$\tilde{\boldsymbol{\epsilon}}_\theta = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, \varnothing) + w\big[\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, \mathbf{c}) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, \varnothing)\big]. \tag{5}$$

We run with $w = 7.5$, the default for ZeroScope checkpoints. Setting w much smaller produces drifting, off-prompt clips; setting it much larger inflates the predicted variance and saturates the output, a well-documented failure mode of plain CFG at large w.

F. Sampling

DDPM uses $\sim$1000 reverse steps and is impractical here; DDIM trims this to 50–100 deterministically [8]; the DPM-Solver multistep method [7] approaches the same quality in 20–40 steps by exploiting the semi-linear structure of the probability-flow ODE. We use the multistep variant at 40 steps as a balance between latency and visible artefacts.

G. Why a 3D denoiser?

Stacked image diffusion models trained per-frame produce visually plausible stills but inconsistent video: lighting flips, foreground identity drifts, and background texture flickers. Lifting the denoiser to 3D, with temporal self-attention layers connecting positions across frames, lets the model commit to a single spatiotemporal hypothesis rather than independently sampling each frame. ZeroScope inherits this design from its parent model.

III. Model Architecture

ZeroScope v2 (576w) [2] is a latent video diffusion model. The diffusion process operates in the latent space of a frozen VAE; this trades a small reconstruction loss for an $8\times$ memory reduction that is what makes the 24-frame, 40-step inference feasible on a T4 GPU at all.

A. Latent pipeline

Each frame is mapped from $576\times320\times3$ pixels to a $72\times40\times4$ latent. A standard CLIP ViT-B/32 text encoder produces the conditioning embedding that enters the U-Net through cross-attention. After T reverse steps, the VAE decoder maps the clean latent volume back to RGB. Decoding is performed in chunks of four frames to fit T4 memory; this is a memory-management detail with no effect on output.

B. 3D U-Net

The denoiser receives a $(B\times F\times C\times H\times W)$ tensor: batch, frames, channels, spatial height, spatial width. Its encoder runs three resolution levels of spatial and temporal downsampling, each composed of residual blocks followed by spatial self-attention and temporal self-attention. A bottleneck adds cross-attention so the text embedding can re-enter at the deepest layer. The decoder mirrors the encoder with skip connections from the corresponding encoder block.

TABLE I
Baseline generation parameters.

Parameter	Value
Model checkpoint	`cerspense/zeroscope_v2_576w`
Scheduler	DPM-Solver Multistep
Frames	24
Output frame rate	8 FPS ($\approx$ 3 s)
Resolution	576$\times$320
Denoising steps	40
Guidance scale w	7.5
Random seed	42
Precision	float16
Decode chunk size	4 frames

C. Three attention modes

The three attention types serve disjoint roles. Spatial self-attention operates on $H\times W$ within a single frame, giving intra-frame structural coherence (the same role it plays in image diffusion). Temporal self-attention operates on F tokens at a fixed (h, w) position, propagating information across frames. Cross-attention injects the CLIP text embedding into every level of the U-Net. Temporal self-attention is the mechanism that, in principle, prevents flicker; Section V returns to what happens when that mechanism is overwhelmed by high-frequency motion.

D. Text conditioning

CLIP ViT-B/32 [6] encodes the prompt to a fixed 77-token context. Conditioning enters at every cross-attention layer of the U-Net. Sinusoidal temporal position encodings, $\mathrm{PE}(t, 2i) = \sin(t/10000^{2i/d})$, $\mathrm{PE}(t, 2i{+}1) = \cos(t/10000^{2i/d})$, mark the position of each frame within the latent volume.

IV. Experimental Setup

A. Hardware and pipeline configuration

All measurements ran on Google Colab with a single NVIDIA T4 GPU (16 GB) at half precision. CPU offload, VAE slicing, and chunked decode were enabled to fit the model in 15 GB of available VRAM. Table I summarises the values that are fixed across every run reported below; they match the configuration shipped in the project notebook [13].

B. Prompt

We characterize the baseline on a single complex prompt — a night-time cyberpunk cityscape with rain, neon, and a moving camera — chosen because it stresses both visual detail and inter-frame motion. The prompt is paired with a fixed negative prompt that discourages still frames, watermarks, washed-out colour and stylised output, mirroring common practice with this checkpoint.

C. Metrics

Four metrics are computed on the decoded RGB frames; their direction of improvement and library origin are stated in Table II. CLIP-SIM averages the cosine similarity between the prompt's CLIP text embedding and each frame's CLIP

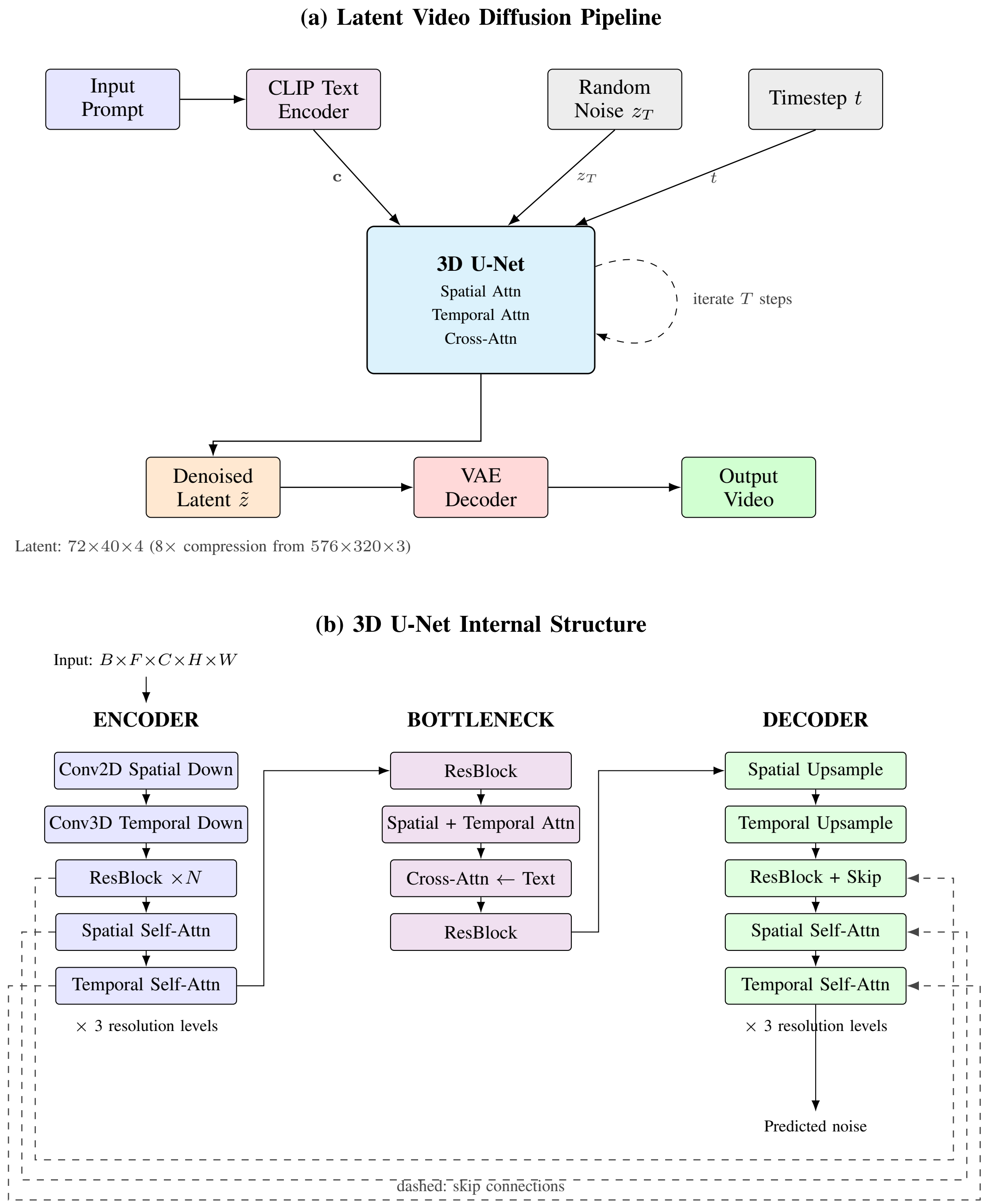

Fig. 1. (a) ZeroScope v2 latent video diffusion pipeline. CLIP encodes the prompt to a text embedding $\mathbf{c}$; the 3D U-Net iteratively denoises a random latent z_T over T DPM-Solver steps, conditioned on $\mathbf{c}$ via cross-attention; the VAE decoder maps the denoised latent to video frames. (b) 3D U-Net internal structure: encoder, bottleneck, decoder. Skip connections (dashed) link corresponding encoder–decoder levels; text embeddings enter via cross-attention in the bottleneck. Drawn from the reference implementation [2].

image embedding, evaluated on the middle 50% of frames (indices $\lfloor n/4 \rfloor$ to $\lfloor 3n/4 \rfloor$) to discount the transitional opening and closing frames; this is the same selection used in the companion code [13].

Temporal LPIPS, SSIM and PSNR are computed between every pair of consecutive frames and averaged. LPIPS uses the AlexNet backbone [11] as is standard for perceptual inter-frame distance in video work; it should be read as a *lower-is-better* measure of perceptual change between adjacent frames.

V. Baseline Measurements and Weakness Analysis

Table III reports the four metrics for the baseline configuration. These values were obtained from the same run that

TABLE II
EVALUATION METRICS.

Metric	Library / model	Dir.
CLIP-SIM	open_clip ViT-B/32 (LAION-2B)	↑
Temporal LPIPS	lpips (AlexNet) [11]	↓
SSIM (pair-wise)	skimage [12]	↑
PSNR (pair-wise)	skimage	↑

TABLE III
BASELINE METRICS ON THE CYBERPUNK PROMPT (SEED 42, 24 FRAMES, 40 STEPS, w=7.5).

Metric	Baseline
CLIP-SIM ↑	0.3241
Temporal LPIPS ↓	0.0687
SSIM ↑	0.6745
PSNR ↑	19.48 dB

underlies Figs. 2–3; they are the reference numbers against which Phase 2 will be compared.

The headline numbers obscure how the metrics behave along the temporal axis. Fig. 3 plots LPIPS and CLIP-SIM frame by frame and reveals two patterns that the averages flatten: the inter-frame perceptual distance stays elevated across the entire clip rather than concentrating at a few transitions, and the per-frame CLIP-SIM swings by ~ 0.03 across the run. We organise the discussion around three weaknesses, each tied to one such pattern.

A. W1 — Inter-frame perceptual drift

Evidence. Temporal LPIPS averages 0.0687 across all consecutive-frame pairs in the baseline. The per-frame trace (Fig. 3, left) hovers in the 0.06–0.075 band rather than spiking at scene cuts, which it would do under a clean motion boundary. **Hypothesis.** The temporal self-attention layers in the 3D U-Net link nearby frames within their receptive footprint, but their effective range across the 24-frame volume is bounded by the attention's locality (only a fraction of frames are co-attended at any one level after downsampling). Under high-frequency content – rain streaks, fast camera motion, fine architectural detail – each local window admits a slightly different sample-level solution. The result is the persistent, low-amplitude drift the LPIPS trace shows. This diagnosis predicts that any intervention that explicitly couples frames – a smoothing post-processor that exploits optical flow, for example – should reduce LPIPS without changing the underlying semantics; this prediction is borne out by the Phase 2 results [14].

B. W2 — Semantic variance and the 77-token bound

Evidence. Per-frame CLIP-SIM in Fig. 3 (right) ranges from ≈ 0.30 to ≈ 0.34 across the 12 evaluated middle frames – a relative spread of $\sim 13\%$ around a mean of 0.3241. **Hypothesis.** The CLIP ViT-B/32 text encoder enforces a hard 77-token context [6]. The detailed prompt combined with our fixed negative prompt crosses this bound under standard tokenisation, so the conditioning vector seen by the U-Net truncates long-tail content silently. The model then samples

TABLE IV
SUMMARY OF THE THREE BASELINE WEAKNESSES, WITH THE INTERVENTION EACH ONE MOTIVATES IN PHASE 2 [14].

ID	Diagnosis	Phase 2 lever
W1	Inter-frame drift	Flow-aware temporal EMA
W2	CLIP token cap	Seed re-rank by CLIP-SIM
W3	Sampler variance	Larger / adaptive step budget

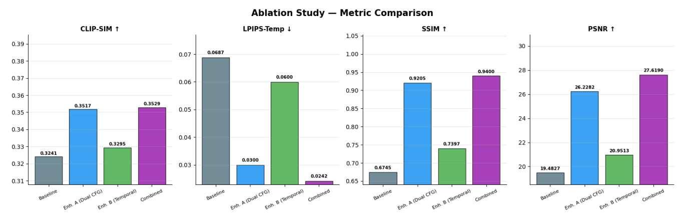

Fig. 2. Per-metric bar chart from the executed pipeline [13]. The leftmost bar (*Baseline*, grey) is the configuration analysed in this paper; the remaining bars preview how the diagnosed weaknesses respond to the Phase 2 interventions [14].

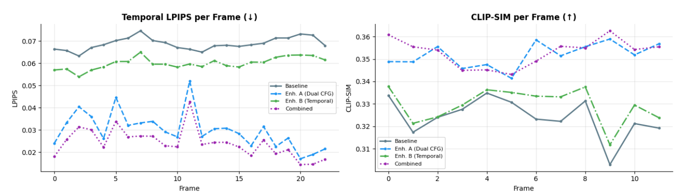

Fig. 3. Per-frame Temporal LPIPS (left, lower is better) and CLIP-SIM (right, higher is better) for the baseline and the Phase 2 runs [13]. The baseline (solid grey) shows the elevated inter-frame drift and the semantic variance discussed in W1–W2.

a spatiotemporal completion that is broadly on-prompt but locally under-constrained, and small drifts in the latent space at later denoising steps move different frames slightly off the prompt manifold. The intervention this diagnosis points to is a selection-based one: regenerate with different seeds and re-rank by CLIP-SIM, which by construction is at least as high as any individual seed and was the basis for the Phase 2 CLIP-reranking module [14].

C. W3 — Sampling-induced inter-frame variance

Evidence. SSIM averaged across consecutive frames is 0.6745 and the average PSNR is 19.48 dB. For reference, a visually static clip would sit near SSIM ≈ 1 and PSNR$\gg 30$ dB, and a baseline this far from that target is consistent with non-trivial residual noise after denoising. **Hypothesis.** The DPM-Solver multistep sampler approaches DDPM-level quality in 20–40 steps for image diffusion, but the same step budget over a 24-frame latent volume distributes its solver error across more positions. Empirically, 40 steps appear to under-denoise the high-frequency components of the latent at certain timesteps, leaving residual variation that decodes to inter-frame structural differences. Increasing steps or switching to an adaptive step controller would test this hypothesis directly; we leave that to follow-on work.

VI. DISCUSSION

The baseline characterized here is a single configuration evaluated on a single complex prompt; the absolute numbers should be read as a reference point for the same pipeline rather than as a benchmark over ZeroScope. The choice to evaluate on the middle 50% of frames for CLIP-SIM is a deliberate noise-control decision and not, on its own, a methodological improvement – it just removes a known source of between-clip variance from the metric. The frame-to-frame SSIM and PSNR measure how similar adjacent frames are, which is what we want for a stability diagnosis but is not a direct measure of generation quality; an artificially still video would score perfectly. Each diagnosis above is internally consistent with one published cause (attention locality, encoder context bound, sampler error), but the mapping from a metric to a root cause is a hypothesis until the predicted intervention is run – which is the role Phase 2 plays.

A second caveat is data scope. We did not vary the random seed for the baseline measurement itself; the per-seed variance that the companion experiment surfaces (CLIP-SIM in $\{0.3241, 0.3264, 0.3517\}$ for seeds 42–44 [13]) is substantial, and a properly bounded baseline would report mean$\pm$std over several seeds. We treat the single-seed reading as the lower bound it implicitly is, and discuss seed sensitivity as direct evidence for W2.

VII. Limitations

Three limitations are worth stating directly. First, the analysis uses a single complex prompt; W1's high LPIPS is partly a function of the scene's intrinsic motion, and a static prompt would lower the floor. Second, the four metrics correlate with but do not coincide with human visual judgement, and a perceptually noticeable artefact can sit under any of them. Third, the inferred root causes (attention range, token truncation, sampler error) are causal hypotheses backed by single-condition evidence, and a stronger attribution would require ablations isolating each factor.

VIII. Conclusion

We presented a baseline characterization of ZeroScope v2 (576w) for short text-to-video generation on commodity hardware: a worked diffusion objective, an attention-level description of the 3D U-Net, and a four-metric snapshot at the operating point we use throughout (24 frames, 40 DPM-Solver steps, w=7.5, seed 42). The pipeline exhibits three concrete weaknesses — inter-frame perceptual drift, semantic variance under CLIP truncation, and inter-frame variance from a fixed sampler budget — and each weakness points to a specific intervention. Those interventions are the subject of the companion Phase 2 work [14], which the per-metric chart in Fig. 2 previews.

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